

International AS and A-level

Biology (9610)

BL03

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL BIOLOGY (9610) BL03 - JANUARY 2025

Students appeared to find the paper accessible and were able to complete it in the time allocated. The skills assessed and demand of the paper were comparable to previous series.

Students are much less likely to use standard phrases for *Evaluate* questions eg 'correlation does not mean causation', and less likely to misuse the phrase 'results due to chance'. This indicated that teachers and/or students are responding to AQA guidance given in reports on the exam or in webinars.

QUESTION 01

01.1

This gap-fill activity proved a good first question on the paper with students scoring very well. The majority of students (over 80%) gained both marks for correct use of biological terms. The most common mistake was *population* instead of *community*.

01.2

Drawing this pyramid of biomass proved challenging with only around a fifth of students gaining both marks. Many students did not follow the instructions to use data from **Table 1** and use a suitable scale. The x-axis label and/or unit was frequently omitted. Some students who did use a suitable scale then drew bars to incorrect widths.

01.3

Although a lot of good answers were seen, a large number of students only gained one mark for the wrong context. When referring to energy losses, these should have been from the trout and not the worm.

01.4

Students found this conversion between units more challenging than expected with less than half gaining both marks for a correct answer given in standard form.

01.5

Less than 10% of students gained the two marks for explaining why the pyramid of numbers for the food chain was 'top heavy'. Many answers were generic statements comparing pyramids of number and biomass without linking the explanation to the food chain in **Figure 1**.

QUESTION 02

02.1

The most common incorrect answer was 'stabilising' with students just looking at the shape of the curve in **Figure 3** and not applying the context of temperature change selecting for smaller and against larger body length. Occasionally students gave the incorrect answer 'disruptive'.

02.2

Most students correctly drew either a skewed curve or a new normal distribution curve to the left of the mean. Despite the instruction to draw the curve of **Figure 3**, many students drew a new graph in the space at the bottom of page 6. This meant it was difficult to compare the new curve with the existing curve for some students. Very occasional incorrect answers included straight lines or s-shaped curves.

02.3

Some students did try to answer this question by describing speciation generically, but they were expected to use the information provided.

QUESTION 03

03.1

The terms 'leaching' and 'eutrophication' are from the specification so students should be familiar with them. Although well answered, common incorrect answers included draining, erosion, accumulation and biomagnification. There were also some spellings of eutrophication that were not credited as they were not phonetically correct.

03.2

Very good answers were seen to this question with logical, step-by-step descriptions of the effects of eutrophication. Many answers included all 4 marking points and almost half the students gained full marks.

03.3

Some answers made good use of all of the data and referred to yield, protein content, and value added. Students were expected to use the data to compare the effects of different masses of fertiliser and not just repeat values from **Table 2**.

QUESTION 04

04.1

Genetic bottleneck is a term from the specification, but many students seemed unfamiliar with its use.

04.2

This was a straightforward calculation that the majority of students (over three quarters) gained the mark for. The most common error was calculating the percentage change in mass.

04.3

This was well answered with almost 60% of students correctly describing that the difference was likely to be significant. It was good to see very few answers referring to 'results due to chance'.

04.4

Many students (over 50%) gained two marks for the ideas of the female seal protecting her own pup and increasing the population. Better students brought in ideas from the rest of the question about the high energy cost to the mother of feeding a pup without hunting for herself, and of genetic similarity within the population.

QUESTION 05

05.1

Students generally answered well with most describing the control of temperature and pH. Common incorrect answers included giving methods of monitoring these variables (thermometer and pH probe) rather than controlling them, or controlling oxygen levels or light intensity.

05.2

A surprising number of wrong answers were seen with students reading from the incorrect end of the liquid (giving an answer of 38) or finding the difference between each end of the liquid droplet (giving an answer of 26). Another common incorrect answer was 10 (not reading from bottom of meniscus).

05.3

Lots of good full-mark answers were seen. However, there were also a lot of 2-mark answers not given to 1 decimal place or with incorrect rounding. Some students rounded too early, losing precision, eg rounding 10.99 to 11.

05.4

The more able students gave the readings that would be needed if calculating RQ using a respirometer. The mark was allowed if students specified that RQ could be calculated from the volume of oxygen consumed and the volume of carbon dioxide produced.

05.5

This was a very well answered question with most students (over 80%) gaining both marks. Occasionally students lost a mark for incorrect rounding, and some students were credited with 1 mark taking correct readings from the graph or a correct calculation from incorrect graph readings.

05.6

Although most students scored at least one mark, usually marking point 1, less than 10% scored the full three marks. This was generally because students did not use information from **Figure 9** but gave generic statements like 'no statistical test' or 'no repeats'. There was no 'do not allow energy produced' for any of the marking points, but the incorrect statement 'energy produced' was seen in a number of responses.

05.7

A lot of very good answers were seen to this use of a graph to compare fructose and lactose. The main reason students did not gain marks was not using data from the graph as they were instructed to. Some students did try to use data but misread the scale(s). Although a tolerance was given for readings, students were expected to be as accurate as possible.

05.8

Many of the answers to this question focused on the monomers glucose and galactose, suggesting that galactose could not be used as a respiratory substrate. However, the graph and previous questions showed that glucose was a very good substrate. If lactose had been hydrolysed, there should be a large volume of carbon dioxide produced.

QUESTION 06

06.1

This 1-mark question assessed students' ability to determine a value from a graph and was very well answered with the majority gaining the mark. Of the incorrect answers, the most common mistake was finding the difference between the initial and final values for crop loss ($76 - 46 = 30$).

06.2

A surprising number of students (over 10%) did not attempt this 3-mark calculation. The three steps required were: reading the values for current and predicted crop loss from the graph, using these values to calculate percentage increase, and dividing the increase by 4 to give the increase per °C of temperature change.

06.3

This question was not answered strongly, with few students directly answering the question which was linking an increase in global temperature to crop loss. Many answers suggested possible effects on insects eg increased range, without linking insects to crop loss. A lot of answers focused on enzymes and changes to rate of reaction or even denaturation, but this did not answer the question.

06.4

Although a lot of students had the right ideas, answers often lacked detail. For example, stating 'no statistical test' or 'only three crops' without linking this to agreeing or disagreeing with the conclusion. There was a high percentage of students who did not attempt this question part.

QUESTION 07

07.1

This was a very well answered question on the role of chlorophyll, with almost 40% of students gaining the full 5 marks. A large percentage of students (over 80%) gained 3 marks, generally marking points 1, 2 and 3, as they omitted the photolysis of water and replacement of the electrons lost from chlorophyll.

Answers were concise and well-structured with good use of key terms and a good understanding of this area of the specification.

A misconception that was seen on a number of scripts was that light energy is converted into electrons.

07.2

Many clear, concise answers were seen to this question on the role of the Krebs cycle in ATP production. It was insufficient to say that the Krebs cycle 'produced ATP' without the phrase or idea of substrate-level phosphorylation for marking point 4. Some answers included unnecessary levels of detail about the role of chemiosmosis in the generation of ATP. Occasional incorrect references to the light-independent reactions of photosynthesis or NADP were seen.

07.3

There were some very good, well-structured and detailed answers to this 6-mark question on how farming practices increase the efficiency of energy transfer. Three quarters of students scored at least 2 marks with almost 10% gaining full marks. The most common points awarded referred to restricting movement and keeping the environment warm. It was less common to see reference to antibiotic and hormone usage. Some students wrote about aspects of the nitrogen cycle such as ploughing without making a link to energy transfer. Some answers included succession which was not relevant to the question.